



程序设计原理

联合体与抽象数据类型

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Unions in C++

DEFINITION

A union is a struct that holds only one of its members at a time during program execution.

EXAMPLE

```
union WeightType
```

```
{
```

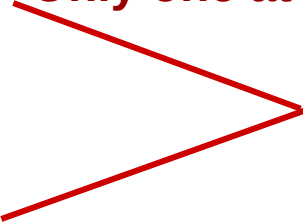
```
    long wtInOunces;
```

```
    int  wtInPounds;
```

```
    float wtInTons;
```

```
};
```

Only one at a time

A red arrow originates from the text "Only one at a time" and points towards the list of members inside the union, emphasizing that only one member can be active at any given time.



Using Unions

```
union WeightType // Declares a union type
{
    long wtInOunces;
    int  wtInPounds;
    float wtInTons;
};

WeightType weight; // Declares a union variable

weight.wtInTons = 4.83;

// Weight in tons is no longer needed
// Reuse the memory space

weight.wtInPounds = 35;
```



Abstraction

- **Abstraction** is the **separation** of the essential qualities of an object from the details of how it works or is composed
- Focuses on **what, not how**
 - Is necessary for managing large, complex software projects



Control Abstraction

- **Control abstraction** separates the logical properties of an action from its implementation

Search (list, item, length, where, found);

- The function call depends on the function's specification (description), not its implementation (algorithm)



Data Abstraction

- **Data abstraction** separates the logical properties of a data type from its implementation

LOGICAL PROPERTIES

What are the possible values?

What operations will be needed?

IMPLEMENTATION

How can this be done in C++?

How can data types be used?



Data Type

set of values
(domain)

allowable operations
on those values

FOR EXAMPLE, data type `int` has

domain

-32768 ... 32767

operations

+, -, *, /, %, >>, <<



Abstract Data Type (ADT)

- An **abstract data type** is a data type whose properties (domain and operations) are specified (*what*) independently of any particular implementation (*how*)

For example . . .



ADT Specification Example

TYPE

Time

DOMAIN

Each Time value is a time in hours, minutes, and seconds.

OPERATIONS

Set the time

Print the time

Increment by one second

Compare 2 times for equality

Determine if one time is “less than” another



Another ADT Specification

TYPE

ComplexNumber

DOMAIN

Each value is an ordered pair of real numbers (a, b) representing $a + bi$

OPERATIONS

Initialize the complex number

Write the complex number

Add

Subtract

Multiply

Divide

Determine the absolute value of a complex number



Several Possible Representations of ADT Time

3 int variables

10

45

27

3 strings

3-elem “10” **array**

“45”

“27”

Choice of representation depends on time, space, and algorithms needed to implement operations

10

45

27



Some Possible Representations of ADT ComplexNumber

struct with 2 float members

-16.2	5.8
.real	.imag

2-element float array

-16.2	5.8
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